

Database Schema

Figure 7.1: Entity Relationship Diagram (ERD) of the Lab Resource Utilization Platform

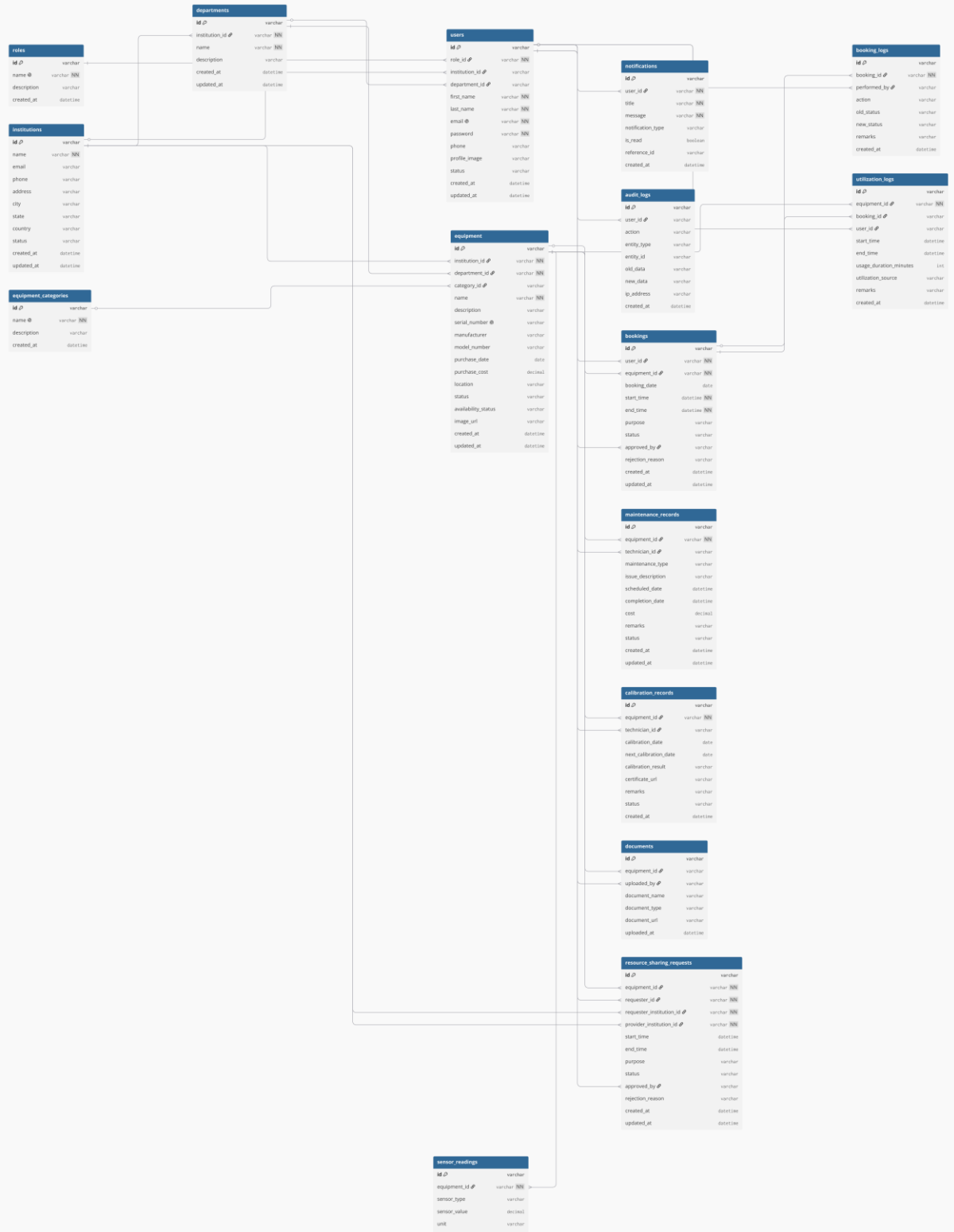
07. Database Schema

7.1 Overview

The Lab Resource Utilization Platform uses a PostgreSQL relational database to efficiently manage laboratory resources, users, institutions, bookings, maintenance, calibration, utilization tracking, and notifications. The database schema is designed using normalization principles to ensure data integrity, reduce redundancy, and maintain efficient relationships between entities.

The schema consists of multiple interconnected tables linked through primary and foreign key relationships. It supports authentication, equipment lifecycle management, booking workflows, resource sharing, maintenance scheduling, utilization monitoring, notifications, and audit logging.

Database Schema Diagram



The above figure illustrates the complete relational database schema showing all entities, primary keys, foreign keys, and relationships used in the Lab Resource Utilization Platform.

7.2 Database Tables

The database is organized into multiple entities, each responsible for storing information related to a specific module of the system.

Table	Description	
Roles	Stores user roles and permissions.	
Institutions	Stores institution information.	
Departments	Stores department details under institutions.	
Users	Stores user accounts and profile information.	
Equipment Categories	Stores categories of laboratory equipment.	
Equipment	Stores equipment inventory and availability.	
Bookings	Stores equipment booking requests.	
Booking Logs	Maintains booking approval history.	
Maintenance Records	Stores maintenance schedules and history.	
Calibration Records	Stores calibration schedules and certificates.	
Resource Sharing Requests	Stores inter-institution equipment sharing requests.	
Utilization Logs	Tracks actual equipment usage duration.	
Sensor Readings	Stores IoT sensor data collected from equipment.	
Notifications	Stores user notifications and alerts.	
Documents	Stores uploaded equipment-related documents.	
Audit Logs	Maintains system activity history.	

7.3 Database Relationships

The ER Diagram illustrates the relationships between all entities.

Major relationships include:

- One Role can be assigned to multiple Users.
- One Institution contains multiple Departments.
- One Department contains multiple Users.
- One Department manages multiple Equipment.
- One Equipment belongs to one Equipment Category.
- One User can create multiple Bookings.
- One Equipment can have multiple Bookings.
- One Equipment can have multiple Maintenance Records.
- One Equipment can have multiple Calibration Records.
- One Equipment can generate multiple Utilization Logs.
- One Equipment can receive multiple Sensor Readings.
- One User can receive multiple Notifications.
- One User can generate multiple Audit Logs.
- One Equipment can have multiple Documents.
- One Equipment can participate in multiple Resource Sharing Requests.

7.4 Key Features of the Database

The database provides the following features:

- Relational PostgreSQL database
- Normalized table structure
- Primary and Foreign Key constraints
- Referential Integrity

- Efficient indexing
- Secure user authentication support
- Complete equipment lifecycle management
- Resource booking and approval tracking
- Maintenance and calibration history
- Equipment utilization monitoring
- Notification management
- Audit logging for system activities

Advantages

The proposed database schema provides several advantages:

- Reduces data redundancy through normalization.
- Ensures consistency using relational constraints.
- Supports multiple institutions and departments.
- Enables efficient equipment management.
- Maintains complete booking history.
- Tracks equipment utilization in real time.
- Supports maintenance and calibration scheduling.
- Provides scalable architecture for future enhancements.

7.6 Conclusion

The database schema forms the backbone of the Lab Resource Utilization Platform by integrating all functional modules into a single relational database. Through well-defined entities and relationships, it enables secure data management, efficient resource allocation, comprehensive utilization tracking, and reliable reporting across multiple institutions.

